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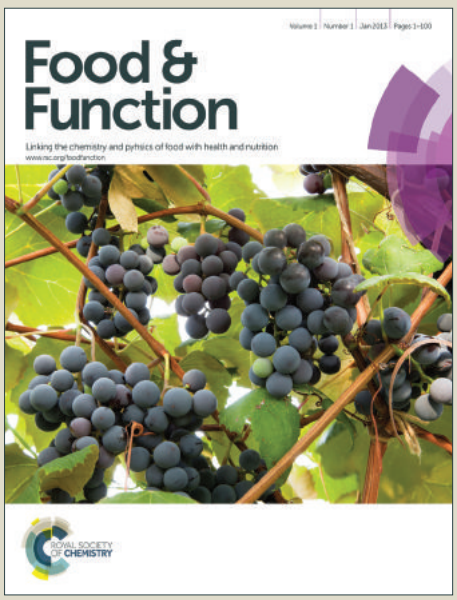

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Two *Ganoderma* species: Profiling of phenolic compounds by HPLC-DAD, antioxidant, antimicrobial and inhibitory activities on key enzymes linked to Diabetes mellitus, Alzheimer's disease and skin disorders

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Abstract

This work reports the antioxidant, antimicrobial, and inhibitory effects of methanol and water extracts from *Ganoderma applanatum* (GAM: methanol extract and GAW: water extract) and *G. resinaceum* (GRM: methanol extract and GRW: water extract) against cholinesterase, tyrosinase, α-amylase and α-glucosidase. Total phenolics, flavonoids contents, and HPLC profile of phenolic components present in the extracts, were also determined. Antioxidant activities were investigated by using different assays, including DPPH, ABTS, FRAP, CUPRAC, phosphomolybdenum and metal chelating assays. Antimicrobial activity of the tested *Ganoderma* extracts was also studied by broth microdilution method. Generally, the highest antioxidant (59.24 mgTEs g^{-1} extract for DPPH, 41.32 mgTEs g^{-1} extract for ABTS, 41.35 mgTEs g^{-1} extract for CUPRAC, 49.68 mgTEs g^{-1} extract for FRAP, 130.57mgAAEs g^{-1} extract for phosphomolybdenum and 26.92 mgEDTAEs g^{-1} extract) and enzyme inhibitory effects (1.47 mgGALAEs g^{-1} extract for AChE, 1.51 mgGALAEs g^{-1} extract for BChE, 13.40 mgKAEs g^{-1} extract for tyrosinase, 1.13 mmolACEs g^{-1} extract for α-amylase and 2.20 mmolACEs g^{-1} extract for α-glucosidase) were observed in GRM, which had the highest concentrations of phenolics (37.32 mg GAEs g^{-1} extract). Again, *Ganoderma* extracts possess weak antibacterial and antifungal activities. Apigenin and protocatechuic acid were determined as the main component in GRM (1761 μg g^{-1} extract) and GAM (165 μg g^{-1} extract), respectively.

The results suggest that the *Ganoderma* species may be considered as a candidate for preparing new food supplements and can represent a good model to the development of new drug formulations.

Keywords: *Ganoderma*; antioxidant; antimicrobial; inhibitory effect; HPLC; Turkey.

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1. Introduction

For millennia, mushrooms are considered valuable source of nutrients and pharmaceuticals. In terms of nutrition, they have little fat and calories, but have higher protein and vitamins.[1] As sources of pharmaceuticals, mushrooms possess a broad spectrum of biological activities such as antioxidant, antimicrobial and anticarcinogenic.[2] In this sense, many mushrooms with both nutritional and pharmacological properties have recently come to the attention of many scientists.[3, 4] They have shown to be a great source for the development of food supplements in industry, and a good model for pharmaceutical field.

The genus *Ganoderma* has used to promote human health since the prehistoric ages. By now, more than 100 *Ganoderma* species were recorded, and the genus is represented by six endemic species *(G. adspersum, G. applanatum, G. carnosum, G. lucidum, G.pfeifferi* and *G. resinaceum*) in Turkey.[5] The *Ganoderma* species is considered one of the most important medicinal mushrooms and is traditionally used in the treatment of various ailments, including cancer, hypertension, gastric ulcer, tumor, kidney and cardiovascular problems.[3, 6-8] In this context, *Ganoderma* species (especially *G. lucidum*) are used to prepare infusions and have been sold as functional supplements at market in worldwide.[9] Based on these informations, recent studies revealed chemical composition and biological properties of some *Ganoderma* species.[10-14] To the best of our knowledge, biological activities of extracts from G. applanatum and G. resinaceum and their chemical composition have not been fully characterized yet.

The objective of this study was to compare the chemical components, the antioxidant, antimicrobial, enzyme inhibitory effects of water and methanol extracts

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from two *Ganoderma* species (*G. applanatum* and *G. resinaceum*). The obtained results may be valuable for preparing new food supplements and can represent a good model to the development of new drug formulations.

2. Materials and methods

2.1. Mushroom materials

Fruiting bodies of two *Ganoderma* mushrooms species were collected at full maturity from different regions of Turkey and were authenticated based on their microscopic and macroscopic characteristics by Dr. Halil Gungor, Department of Biology, Mugla University, Mugla, Turkey. They were stored at Mugla University Science Faculty Herbarium Laboratory. The fruiting bodies of two *Ganoderma* species were cut into small parts and then hot air-dried in an oven for 48 h at 40 °C before further analysis.

Localities and collection periods of *Ganoderma* species are as following:

1. *Ganoderma resinaceum* Boud.: Köyceğiz, Muğla-Turkey, in *Liquidambar orientalis* Mill. forest, on *L. orientalis* stump, 25 October, 2013; (Voucher No: H 618-a)
2. *Ganoderma applanatum* (Pers.) Pat.: Düzce-Turkey, Balıklı lake plateau, *Fagus* sp.-*Abies* sp. mixed forest, on rottening *Fagus* sp. stump, 2 November, 2013, (Voucher No: H 629).

2.2. Preparation of extracts

For methanol extract, hot air-dried mushroom materials (10 g) were macerated with 100 mL of methanol at room temperature for 24 h. For water extract, hot air-dried mushroom materials (10 g) were extracted by boiling deionized water (100 mL) for 15 min. Methanol was then removed with a rotary evaporator at 40 °C. The water extract was freeze–dried. All extracts were stored at +4 °C until analyses. Yields of methanol and

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water extracts were determined as 18.90 and 8.13% (w/w) for *Ganoderma applanatum,* and 4.85 and 2.88% (w/w) for *G. resinaceum,* respectively.

2.3. Phenolic compounds by reversed-phase high-performance liquid chromatography (RP-HPLC)

Phenolic compounds were evaluated by RP-HPLC (Shimadzu Scientific Instruments, Tokyo, Japan). Detection and quantification were carried out with a LC-10ADvp pump, a Diode Array Detector, a CTO-10Avp column heater, SCL-10Avp system controller, DGU-14A degasser and SIL-10ADvp auto sampler (Shimadzu Scientific Instruments, Columbia, MD). Separations were carried out at 30 °C on Agilent® Eclipse XDB C-18 reversed-phase column (250 mm x4.6 mm length, 5 μm particle size). The eluates were detected at 280 nm.

Phenolic compositions of the extracts were determined by a modified method of Sarikurkcu *et al.*[15] Protocatechuic acid, (+)-catechin, p-hydroxybenzoic acid, chlorogenic acid, caffeic acid, (-)-epicatechin, ferulic acid, benzoic acid, rutin, rosmarinic acid and apigenin were used as standards. The phenolic compounds were characterized according to the UV spectra, retention times, and comparison with authentic standards. The identified phenolic acids were quantified by comparison of the area of their peaks recorded at 280 nm with calibration curves obtained from commercial standards of each compound. The results were expressed in mg per g of extract.

2.4. Total phenolic and flavonoids content

Total phenolics and flavonoids contents were determined by employing the methods given in literature.[16]

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(a) Total phenolics content. Sample solution (0.25 mL) was mixed with diluted Folin-Ciocalteu reagent (1 mL, 1:9) and shaken vigorously. After 3 minutes, Na_2CO_3 solution (0.75 mL, 1%) was added and the sample absorbance was read at 760 nm after incubation for 2 hours at room temperature. Total phenolics content was expressed as equivalents of gallic acid.

(b) Total flavonoids content. Briefly, sample solution (1 mL) was mixed with the same volume of aluminium trichloride (2%) in methanol. Similarly, a blank was prepared by adding sample solution (1 mL) to methanol (1 mL) without $AlCl_3$. The sample and blank absorbance were read at 415 nm after incubation for 10 minutes at room temperature. The absorbance of blank was subtracted from that of the sample. Total flavonoids content was expressed as equivalents of rutin.

2.5. Total antioxidant activity by phosphomolybdenum method

Total antioxidant activities were evaluated using phosphomolybdenum method as previously described in literature.[16] Sample solution (0.2 mL) was combined with 2 mL of reagent solution (0.6 M sulfuric acid, 28 mM sodium phosphate and 4 mM ammonium molybdate). The sample absorbance was read at 695 nm after incubation for 90 minutes at 95 °C. Total antioxidant capacity was expressed as equivalents of trolox.

2.6. Radical scavenging activity

The effects of the samples on 1,1-diphenyl-2-picrylhydrazyl (DPPH)[17] and ABTS [2,2 Azino-bis (3-ethylbenzothiazloine-6-sulfonic acid)] cation[18] radicals were estimated according to the procedure in literature.

(a) Free radical scavenging activity (DPPH). Sample solution (1 mL) was added to a 4 mL of 0.004% methanol solution of DPPH. The sample absorbance was read at 517

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nm after incubation for 30 minutes at room temperature in dark. DPPH scavenging activity was expressed as equivalents of trolox.

(b) ABTS (2,2 Azino-bis (3-ethylbenzothiazloine-6-sulfonic acid)) radical cation scavenging activity. Briefly, $ABTS^{\cdot+}$ radical cation was produced directly by reacting 7 mM ABTS solution with 2.45 mM potassium persulfate and allowing the mixture to stand for 12-16 in dark at the room temperature. Prior to beginning the assay, ABTS solution was diluted with methanol to obtain an absorbance of 0.700±0.02 at 734 nm. Then, sample solution (1 mL) was added to ABTS solution (2 mL) and mixed. The sample absorbance was read at 734 nm after incubation for 30 minutes at room temperature. ABTS radical cation scavenging activity was expressed as equivalents of trolox.

2.7. Reducing power

The reducing power was investigated using cupric ion reducing (CUPRAC) and ferric reducing antioxidant power (FRAP) methods, as previously described in literature.[16]

(a) Cupric ion reducing (CUPRAC) method. Sample solution (0.5 mL) was added to premixed reaction mixture containing $CuCl_2$ (1 mL, 10 mM), neocuproine (1 mL, 7.5 mM) and NH_4Ac buffer (1 mL, 1 M, pH 7.0). Similarly, a blank was prepared by adding sample solution (0.5 mL) to premixed reaction mixture (3 mL) without $CuCl_2$. Then, the sample and blank absorbance were read at 450 nm after incubation for 30 minutes at room temperature. The absorbance of the blank was subtracted from that of the sample. CUPRAC activity was expressed as equivalents of trolox.

(b) Ferric reducing antioxidant power (FRAP) method. Sample solution (0.1 mL) was added to premixed FRAP reagent (2 mL) containing acetate buffer (0.3 M, pH

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3.6), 2,4,6-tris(2-pyridyl)-s-triazine (TPTZ) (10 mM) in 40 mM HCl and ferric chloride (20 mM) in a ratio of 10:1:1 (*v: v: v*). Then, the sample absorbance was read at 593 nm after incubation for 30 minutes at room temperature. FRAP activity was expressed as equivalents of trolox.

2.8. Metal chelating activity on ferrous ions

Metal chelating activity on ferrous ions was evaluated by the method described by Zengin *et al.*[16] Briefly, sample solution (2 mL) was added to $FeCl_2$ solution (0.05 mL, 2 mM). The reaction was initiated by the addition of 5 mM ferrozine (0.2 mL). Similarly, a blank was prepared by adding sample solution (2 mL) to $FeCl_2$ solution (0.05 mL, 2 mM) and water (0.2 mL) without ferrozine. Then, the sample and blank absorbance were read at 562 nm after incubation for 10 minutes at room temperature. The absorbance of blank was subtracted from that of the sample. The metal chelating activity was expressed as equivalents of EDTA.

2.9. Enzyme inhibitory activity

Enzyme inhibitory activities the samples were determined using cholinesterase (ChE), α-amylase, α-glucosidase and tyrosinase enzymes by employing the methods given in the literature.[18]

(a) Cholinesterase inhibition. Sample solution (50 μL) was mixed with DTNB (125 μL) and AChE (or BChE) solution (25 μL) in Tris-HCl buffer (pH 8.0) in a 96-well micro plate and incubated for 15 minutes at 25 °C. The reaction was then initiated with the addition of acetyl thiocholine iodide (ATCI) or butyrylthiocholine chloride (BTCl) (25 μL). Similarly, a blank was prepared by adding sample solution to all reaction reagents without enzyme (AChE or BChE) solution. The sample and blank absorbance were read at 405 nm after incubation for 10 minutes at 25 °C. The absorbance of blank

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was subtracted from that of the sample and the cholinesterase inhibitory activity was expressed as equivalents of galanthamine.

(b) α-Amylase inhibition. Sample solution (25 μL) was mixed with α-amylase solution (50 μL) in phosphate buffer (pH 6.9 with 6 mM sodium chloride) in a 96-well micro plate and incubated for 10 minutes at 37 °C. After pre-incubation, the reaction was initiated with the addition of starch solution (50 μL, 0.05%). Similarly, a blank was prepared by adding sample solution to all reaction reagents without enzyme (α-amylase) solution. The reaction mixture was incubated for 10 minutes at 37 °C. The reaction was then stopped with the addition of HCl (25 μL, 1 M). This was followed by the addition of iodine-potassium iodide solution (100 μL). The sample and blank absorbance were read at 630 nm. The absorbance of blank was subtracted from that of the sample and the α-amylase inhibitory activity was expressed as equivalents of acarbose.

(c) α-Glucosidase inhibition. Sample solution (50 μL) was mixed with glutathione (50 μL) and α-glucosidase solution (50 μL) in phosphate buffer (pH 6.8) and PNPG (50 μL) in a 96-well micro plate and incubated for 15 minutes at 37 °C. Similarly, a blank was prepared by adding sample solution to all reaction reagents without enzyme (α-Glucosidase) solution. The reaction was then stopped with the addition of sodium carbonate (50 μL, 0.2 M). The sample and blank absorbance were read at 400 nm. The absorbance of blank was subtracted from that of the sample and α-glucosidase inhibitory activity was expressed as equivalents of acarbose.

(d) Tyrosinase inhibition. Sample solution (25 μL) was mixed with tyrosinase solution (40 μL) and phosphate buffer (100 μL, pH 6.8) in a 96-well micro plate and incubated for 15 minutes at 25 °C. The reaction was then initiated with the

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addition of L-DOPA (40 μL). Similarly, a blank was prepared by adding sample solution to all reaction reagents without enzyme (tyrosinase) solution. The sample and blank absorbance were read at 492 nm after incubation for 10 minutes at 25 °C. The absorbance of blank was subtracted from that of the sample and tyrosinase inhibitory activity was expressed as equivalents of kojic acid.

2.10. Determination of antimicrobial activity

(a) Microorganisms. The *in vitro* antibacterial activities of the *Ganoderma* extracts were evaluated against a total of eight standard bacteria which were human pathogen and include *Escherichia coli* ATCC 25922, *Pseudomonas aeruginosa* ATCC 15442, methicillin sensitive *Staphhyloccocus aureus* ATCC 25623, *Klebsiella pneumoniae* 70603 (MSSA), methicillin resistant *Staphhyloccocus aureus* ATCC 43300 (MRSA), *Salmonella enteritidis* ATCC 13076, *Streptococcus pneumoniae* ATCC 10015 and *Sarcina lutea* ATCC 9341. Also antifungal activities were determined by using *Candida albicans* and *Candida parasilopsis* yeast isolates. In addition, fourteen methicillin resistant *S. aureus* strains isolated from clinical samples were used to determine anti-MRSA activity of the *Ganoderma* extracts. Agar screening, oxacillin disc diffusion and broth microdilution methods determined the methicillin resistances of strains. The standard microorganisms and yeast were got access to from Microbiology Research Laboratory, Department of Biology, Selcuk University, and Konya, Turkey.

(b) Preparation of bacterial suspensions. The bacterial and yeast suspensions were standardized following the Clinical and Laboratory Standards Institute (CLSI) guidelines for aerobic bacteria.[19] All of the tested microorganisms were grown in Brain Heart Infusion Broth (BHIB) for 18–24 h, followed by the adjusting of bacterial suspension to the turbidity equivalent to 0.5 McFarland solutions (1×10^8 CFU mL^{-1}) in

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the sterile physiological water tubes. Then 0.5 McFarland tubes diluted to 5×10^5 CFU mL^{-1} for the antimicrobial activity testing.

(c) Determination of minimum inhibitory concentrations (MICs). A total of 22 bacterial and two yeast strains were subjected to antimicrobial susceptibility by broth microdilution method according to CLSI[19] with slight modifications.[20] The crude extracts were dissolved in 10% dimethylsulfoxide (DMSO) and the concentrations of the *Ganoderma* extracts tested ranged from 2500 to 1.22 μg mL^{-1}. In broth microdilution assay, sterile U-bottom 96-well plates were loaded with 100 μL of Mueller Hinton Broth and 100 μL of extracts were added into the first wells. Two-fold dilutions of extracts were prepared by transferring 100 μL of medium and extract mix into 11 consecutive well. Then 100 μL of bacterial inoculums were added to each well. Total volume of the assay system in each well was kept 200 μL. Plates without any plant extracts were served as growth control. The assay plates were incubated at 37 °C for 18-24 h. Yeast plates were exposed to 37 °C for 48 h. Gentamicin and Oxacilin were used as positive control. After incubation period microbial growth was determined by adding 20 μL of 2,3,5-Triphenyl-tetrazolium chloride (0.5%) to each well and incubating for 30 minute at 37 °C.[20] The highest dilution of extract that showed no turbidity in broth microdilution assay was considered as MIC.

2.11. Statistical analysis

All the assays were carried out in triplicate. The results were expressed as mean values and standard deviations (mean ± SD). Statistical differences between the extracts were analyzed using Student t-test (α=0.05). All the analyses were carried out using SPSS v14.0 software.

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3. Results and Discussion

3.1. Total phenolics and flavonoids contents

Phenolic compounds have attracted considerable interest due to their biological properties including anti-oxidant, anti-mutagenic and anti-carcinogenic effects. Total phenolics and flavonoids contents in *Ganoderma* extracts were determined in this work. The components were expressed as mg of gallic acid (mg GAEs g^{-1} extract) and rutin (mg REs g^{-1} extract) per g dry extract, respectively. Table 1 shows the phenolics and flavonoids content of the tested extracts. *G. resinaceum* extracts had the highest total phenolic content (37.32 and 36.39 mg GAEs g^{-1} extract for methanol and water extracts, respectively). GRW had higher amounts of total phenolics in comparison with GAW by approximately 1.6-fold. Both methanol extracts have also higher phenolic contents than water extracts. As to total flavonoids content, GRW gave the highest yields of total flavonoids (20.61 mg REs g^{-1} extract) followed by GAM (19.52 mg REs g^{-1} extract), GAW (14.78 mg REs g^{-1} extract) and GRM (13.09 mg REs g^{-1} extract). From these results, it can be concluded that flavonoids are the majority of the total phenolic content in the tested extracts. The total phenolics contents were reported in the literature for other *Ganoderma* species such as *G. atrum* (75.80 mg GAEs g^{-1} of an ethanolic extract, reported by Li *et al.*[21]), *G. australe* (88.3 mg GAEs g^{-1} of an ethanol extract, reported by Bang *et al.*[22]) and *G. lucidum* (63.51 mg GAEs g^{-1} of an water extract, reported by Abdullah *et al.*[23]).

3.2. Free radical scavenging activities by DPPH and ABTS assays

Antioxidant compounds interact with free radicals (DPPH and ABTS) by electron or hydrogen atom transfer, thus converting them into stable and non-radical molecules.

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Table 1 illustrates the DPPH and ABTS radical scavenging ability of *Ganoderma* extracts. The highest radical scavenging activity was demonstrated by GRM (59.24 and 41.32 mg TEs g^{-1} extract in DPPH and ABTS assays, respectively). The radical scavenging activity in both DPPH and ABTS was found to be in the order of GRM>GRW>GAW>GAM. The results showed the similarity of both methods (DPPH and ABTS) occurring one electron-transfer. Generally, the radical scavenging activities followed a similar trend as total phenolic content. These findings are in agreement with Stankovic *et al.*[24] and Pacifico *et al.*[25] They observed strong correlation between total phenolic content and free radical scavenging activity. Several studies reported that *Ganoderma* species had considerable free radical scavenging activities.[21, 26-29]

3.3. Reducing power

Antioxidant properties and reducing power are strongly associated. Reductones have shown to exert antioxidant action by breaking free radical chain reaction by donating a hydrogen atom and thus inhibiting lipid peroxidation. Reducing power was evaluated by the transformation of Fe^{3+} to Fe^{2+} and Cu^{2+} to Cu^{+} in the presence of the tested extracts. The results expressed as trolox equivalents per g of crude extract (TEs g^{-1} extract) are shown in Table 1. The highest FRAP value was found in GRW (55.02 mg TEs g^{-1} extract) and the lowest in GAM (36.40 mg TEs g^{-1} extract). Generally, the tested water extracts had higher ferric reducing ability than methanol extracts. Interestingly, methanol extracts were exhibited higher cupric reducing ability as compared to water extracts. As reported in Table 1, the order of the cupric reducing power activity was GRM>GRW>GAM>GAW. Also, the same order was observed for total phenolic content of tested extracts. Similarly, the results were supported by several researchers, which reported a good correlation between reductive potential and total phenolic content of

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some plant extracts.[15, 30] In the previously published papers, there is only little information about reducing power of *Ganoderma* species.[23, 27, 28, 31]

3.4. Phosphomolybdenum and metal chelating assays

The phosphomolybdenum assay is based on the reduction of Mo(VI) to Mo(V) by antioxidant compounds, especially phenolics. Then, a green phosphate/Mo(V) complex occurs and the complex has maximal absorption at 695 nm. The activity followed a similar pattern as the total phenolics content, GRM (130.57 mg AAEs g^{-1} extract) had the highest followed by GRW (89.20 mg AAEs g^{-1} extract), GAM (84.54 mg AAEs g^{-1} extract) and GAW (25.58 mg AAEs g^{-1} extract), as reported in Table 1. The results revealed that the antioxidant activity of GRM was 1.5-fold higher than that on GAM. From these results, the activities obtained from phosphomolybdenum method could be explained as the presence of phenolics. The strong relationship observed in this study in agreement with other reports in the literature.[31, 32]

Chelating ferrous ion may prevent production of reactive oxygen species because ferrous ion is known to react with hydrogen peroxide, producing highly reactive hydroxyl radicals (Fenton reaction).[33] Therefore, it was considered important to characterize the *Ganoderma* extracts ability to chelate on ferrous ion. In this assay, ferrozine forms a complex with free ferrous ion but not ferrous ion already chelated by a compound (chelator). The ferrozine-ferrous ion complex forms a red color that can be monitored spectrophotometrically at 562 nm. As in phosphomolybdenum method, GRM had highest chelating ability and the results are presented in Table 1. The methanol extracts presented more significant chelating ability than the water extracts, which can explain by the higher amount of phenolics. GAW, whose total phenolic content was lowest, also ranked last on the chelating activity (7.46 mg EDTAEs g^{-1} extract). These

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findings support that chelation of transition metals is one of the ways polyphenols exert their antioxidant activity. Several results about chelating activities of *Ganoderma* species were found in the literature. For example, Mau *et al.*[27] and Kozarski *et al.*[13] found that the extracts of *Ganoderma* species were good chelators for ferrous ion.

3.5. Enzyme inhibitory effect

In recent decades there has been an increase in the occurrence of major health problems including Alzheimer's disease (AD), diabetes mellitus (DM) and skin disorders (SD). For example, AD affects approximately 36 million people in 2010, and it is estimated that this number may increase to 66 million by 2030.[34] Again, DM currently affects more than 300 million people worldwide and is expected to affect about 500 million by 2030.[35, 36] In this context, investigation of different strategies to teat mentioned diseases is one of main topics in scientific world. The key enzymes inhibition is currently the most established strategy for the treatment of AD and DM. At this point, several enzyme inhibitors (tacrine and galathamine for AD, acarbose and viglibose for DM and kojic acid for SD) are used to alleviate the symptoms related with AD and DM. Nevertheless, some of those inhibitors have side effects including gastrointestinal disturbance and hepatotoxic.[37-39] Taking into account these concerns, it is very important to find out safe and novel inhibitors derived from natural origin.

In this work were screened *Ganoderma* extracts against the enzymes AChE, BChE, α-amylase, α-glucosidase and tyrosinase (Table 1). Generally, methanol extracts exhibited the higher enzyme inhibitory activities than water extracts. Similarly, phenolics were present in higher amounts in the methanol extracts. From these results, the enzyme inhibitory of *Ganoderma* species may be partially correlated with total phenolics. All extracts demonstrated inhibitory effect against AChE (062-1.58 mg

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GALAEs g^{-1} extract), BChE (0.70-2.94 mg GALAEs g^{-1} extract), and α-glucosidase (0.53-2.20 mg ACEs g^{-1} extract). Interestingly, the water extracts tested in this study did not have any effects on tyrosinase and α-amylase. However, the highest AChE inhibition was obtained with GAW (1.58 mg GALAEs g^{-1} extract), which also had also the lowest capacity to inhibit BChE (0.70 mg GALAEs g^{-1} extract). As far as we know, this is the first study that reports the enzyme inhibitory activities of *G. applanatum* and *G. resinaceum*.

3.6. Antimicrobial activity

According to the data obtained from broth microdilution test, the most of the strains were resistant to *Ganoderma* extracts and they did not reveal any antimicrobial activity against tester strains. Only water and methanol extracts of *G.applanatum* manifested weak antibacterial activity against *E. coli* at a concentration of 2.5 mg mL^{-1}, and GAM revealed impotent antibacterial activity against *Pseudomonas aeruginosa* (2.5 mg mL^{-1}). In addition to this, GAM was found to be antifungal against *Candida albicans* at a dose of 2.5 mg mL^{-1}, while GAW had no antifungal capacity. This MIC values can be considered as weak antifungal activity. Both the methanol and water extract of this mushroom was poorly effective against *Candida parasilopsis* (2.5 mg mL^{-1}.). Based on the results, it was determined that water and methanol extracts of *G. applanatum* manifested weak antibacterial activity against some Gram-negative bacteria. It was seen that most of the bacteria tested in this study were found to be resistant to this extract.

Especially, anti-MRSA activity was not observed against to clinical isolates. When the *Ganodermaresinaceum* extracts were evaluated, GRM and GRW exhibited slightly antifungal activity against *C. parasilopsis* at a concentration of 2.5 mg mL^{-1} and 1.25 mg mL^{-1}, respectively. This extracts were ineffective against all of the standard bacteria and clinical isolates, which were tested in this study.

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A lot of studies about biological and antimicrobial activities of *Ganoderma* species were reported.[10, 12] In a studies conducted by Kim *et al.*[40], and Ćilerdžić *et al.*[41], it was observed that *G. lucidum* extracts had antibacterial capacity against Gram-positive microorganisms and *E. coli* and *Salmonella typhimurium*. Coletto and Mondino[42] noted that GRM was effective against *Bacillus subtilis* and *Staphylococcus aureus*. Similarly, Li *et al.*[21]reported that *G. atrum* extract revealed antibacterial activity against tested bacteria and MIC values were in the range of 1.5-50 mg mL^{-1}. In our study GAM was effective against *E. coli*, *P. aeruginosa*, *Candida albicans*, and *Candida parasilopsis* at a concentration of 2.5 mg mL^{-1} doses and GRW and GRM extract had slightly antifungal capacity against only *C. parasilopsis*at doses of 1.25 and 2.5 mg mL^{-1}, respectively. Our findings were consistent with the results of the researchers mentioned above.

Although Heleno *et al.*[43]reported that *P. aeruginosa* was resistant to *G. lucidum* extract; in our study *P. aeruginosa* was affected from GAM at adose of 2.5 mg mL^{-1}. This MIC values can be considered as weak antibacterial activity. Mothana *et al.*[44]revealed that *G. pfeifferi* was effective against MRSA strains. Ameri *et al.*[45] found that chloroform and methanol extracts of *G. praelongum, G. resinaceum*and *G. lucidum* manifested weak to moderate anti-MRSA activity against tested strains. But in our study clinical isolates of MRSA strains were resistant to all the extracts tested.

3.7. Phenolics composition

The identification and quantification of phenolic components in *Ganoderma* extracts was determined HPLC-DAD system. To validate the HPLC method for phenolics quantification was performed linearity, LOD (limit of detection), and LOQ (limit of quantification). The results are listed in Table 2.

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The phenolic compounds were characterized and quantified according to the procedure reported in section 2.3.

Eleven standard compounds were analyzed and 10 of them were detected in the tested extracts. Rutin was not found in any of the four extracts. These data highlight that *Ganoderma* extracts present different phenolic profiles (Figure 1). Apigenin, benzoic acid, catechin and epicatechin represented the major compounds in both methanol and water extracts obtained from *G. resinaceum*. As to *G. applanatum* extracts, predominant phenolics were protocatechuic, catechin and ferulic acids. The *p*-hyroxybenzoic, chlorogenic, caffeic, and rosmarinic acids were in minute quantities. Among the studied extracts, the highest level of apigenin was detected in GRM. Also, the extract possess higher amount of catechin and epicatechin level. However, GAM did not contain detectable apigenin. Apigenin, catechin and epicatechin acid have attracted considerable attention due to their reported health benefits. For example, they have been reported to have a strong biological effects including antioxidant, enzyme inhibitory, antimicrobial and anticarcinogenic.[46-51]

In this direction, the strongest antioxidant and enzyme inhibitory effects for observed GRM may be explained as the presence of above-mentioned compounds. As a result, *G. resinaceum* may be recommended as an effective source for food supplements. Similarly, the presence of flavonoids for several mushroom species was observed in the literature.[52-56] Literature is scarce about the levels of phenolic compounds in *Ganoderma* species (*G. lucidum*).[26] Nevertheless, identification and quantification of phenolic components in tested *Ganoderma* extracts have not been previously reported in the literature.

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This study is the first to be reported in the literature regarding the composition of phenolics in *G. applanatum* and *G. resinacaeum* extracts.

4. Conclusion

The study reported herein describes, for the first time, the identification of phenolic components of two *Ganoderma* species, as well as their antioxidant, antimicrobial, and enzyme inhibitory activities. Generally, GRM exhibited the highest antioxidant and enzyme inhibitory activity. In addition, *Ganoderma* extracts manifested antibacterial and antifungal activities against some microorganisms such as *E. coli* and *P. aeruginosa* and *Candida* species, which were human pathogen. These activities are weak as expected for food.

Apigenin and protocatechuic acid were identified as the major phenolic components in GRM and GAM, respectively. Our results suggest possible application of *Ganoderma* species to pharmaceutical and food industries (as functional foods). In this sense, the *Ganoderma* species may be interesting candidate for evaluation in further biological assays including toxicological and *in vivo* assays.

Conflict of interest

The authors have no conflicts of interest.

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Table 1 Total phenolics and flavonoids, radical scavenging, reducing power, total antioxidant, metal chelating and enzyme inhibitory activities of methanol and water extracts from two *Ganoderma* species (mean±SD)*

Assays	*Ganoderma applanatum*		*Ganoderma resinaceum*	
	Methanol	Water	Methanol	Water
Total phenolics (mg GAEs/g extract) **	29.66±0.65aB	21.97±1.36bY	37.32±0.98aA	36.39±1.20aX
Total flavonoids (mg REs/g extract) ***	19.52±0.24aA	14.78±0.22bY	13.09±0.25bB	20.61±0.41aX
DPPH (mg TEs/g extract)****	17.01±0.84bB	33.36±1.20aY	59.24±0.53aA	56.39±0.76bX
ABTS cation (mg TEs/g extract)****	14.85±1.31bB	22.06±0.44aY	41.32±0.39aA	33.40±0.63bX
CUPRAC (mg TEs/g extract)****	27.83±1.01aB	22.03±0.61bY	41.35±0.60aA	36.07±0.74bX
FRAP (mg TEs/ extract)****	36.40±0.23aB	37.61±1.60aY	49.68±0.43bA	55.02±2.03aX
Phosphomolybdenum (mg AAEs/g extract)*****	84.54±0.07aB	25.58±0.55bY	130.57±7.88aA	89.20±3.91bX
Chelating effect (mg EDTAEs/g extract)******	11.97±0.10aB	7.46±0.47bY	26.92±0.14aA	25.44±0.14bX
Acetyl cholinesterase (mg GALAEs/g extract)*******	1.45±0.01bA	1.58±0.02aX	1.47±0.01aA	0.62±0.01bY
Butyryl cholinesterase (mg GALAEs/g extract)*******	2.94±0.19aA	0.70±0.32bX	1.51±0.04aB	1.35±0.12aX
Tyrosinase (mg KAEs/g extract)********	8.47±0.06B	na*********	13.40±0.57A	na
α-Amylase (mmol ACEs/g extract)*********	1.40±0.26A	na	1.13±0.01A	na
α-Glucosidase (mmol ACEs/g extract)*********	2.05±0.01aA	0.53±0.25bX	2.20±0.17aA	0.78±0.19bX

* Different letters (a,b) in the same row indicate significant difference for each mushrooms ($p< 0.05$). Different letters (A,B and X,Y) in the same row indicate significant difference between methanol and water extracts, respectively ($p< 0.05$).
**GAEs, gallic acid equivalents.
***REs, rutin equivalents.
**** TEs, trolox equivalents.
***** AAEs, ascorbic acid equivalents.
****** EDTAEs, disodium edetate equivalents.
******* GALAEs, galanthamine equivalents.
******** KAEs, kojic acid equivalents.
********* ACEs, acarbose equivalents.
********* na, not active

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Table 2 Analytical characteristics for determination of phenolics and amounts (μg g^{-1} extract) of selected phytochemicals in two *Ganoderma* species (mean ± SD) *

No	Phenolic components	*Ganoderma applanatum*		*Ganoderma resinaceum*		Analytical characteristics			
		Methanol	Water	Methanol	Water	Linear range (mg L^{-1})	R^2	LOD (mg L^{-1})	LOQ (mg L^{-1})
1	Protocatechuic acid	165±1 bA	554±5 a	77±2 B	nd**	0.20-25.0	0.9991	0.086	0.260
2	(+)-Catechin	76±1 bB	266±35 aX	287±36 aA	128±3 bY	0.90-113	0.9988	0.172	0.522
3	*p*-Hydroxybenzoic acid	nd	nd	186±2 a	12±1 b	0.20-25.0	0.9994	0.007	0.020
4	Chlorogenic acid	17±1 aB	16±1 aY	51±4 aA	48±2 aX	0.35-45.0	0.9988	0.080	0.241
5	Caffeic acid	nd	nd	65±5 a	60±8 a	0.16-21.0	0.9993	0.054	0.162
6	(-)-Epicatechin	nd	nd	277±15	nd	0.50-66.0	0.9990	0.170	0.514
7	Ferulic acid	58±3 b	103±5 a	nd	nd	0.12-17.0	0.9993	0.004	0.011
8	Benzoic acid	nd	nd	376±12 b	796±20 a	0.85-55.0	0.9998	0.111	0.335
9	Rutin	nd	nd	nd	nd	0.40-55.0	0,9990	0.180	0.550
10	Rosmarinic acid	nd	nd	12±1	nd	0.40-55.1	0.9989	0.050	0.150
11	Apigenin	nd	86±2 Y	1761±15 b	388±4 aX	0.17-11.0	0.9997	0.034	0.104

* Different letters (a,b) in the same row indicate significant difference for each mushrooms (p< 0.05). Different letters (A,B and X,Y) in the same row indicate significant difference between methanol and water extracts, respectively (p< 0.05).

**nd, not detected.

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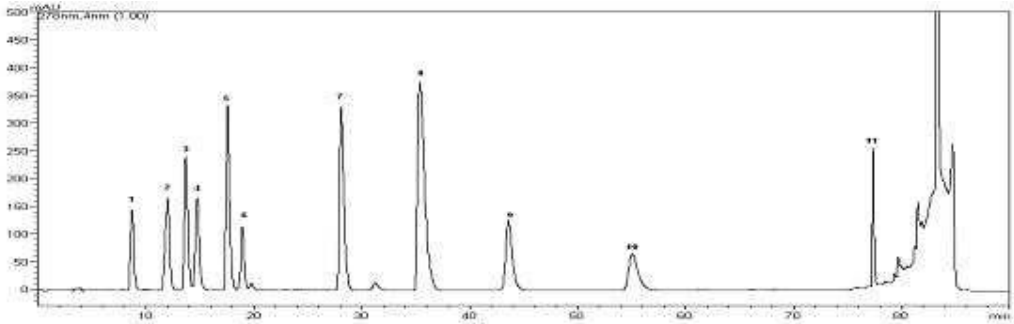

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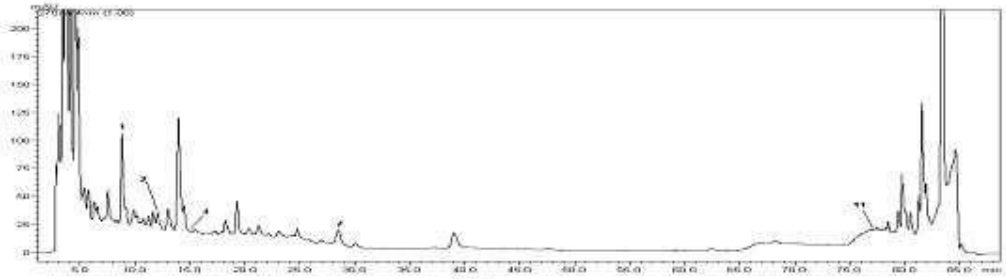

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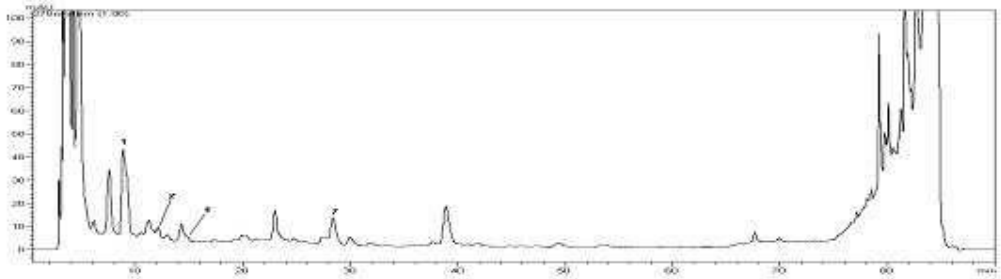

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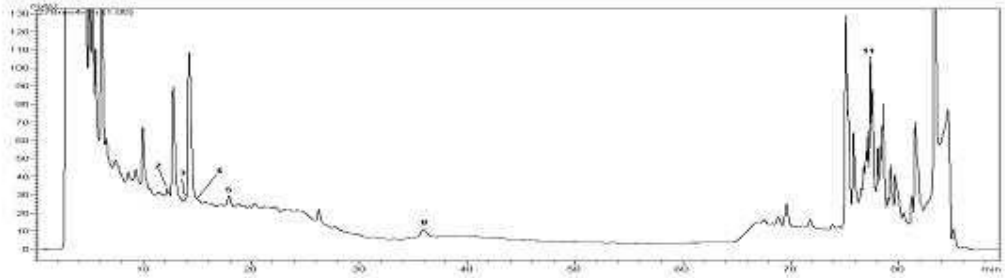

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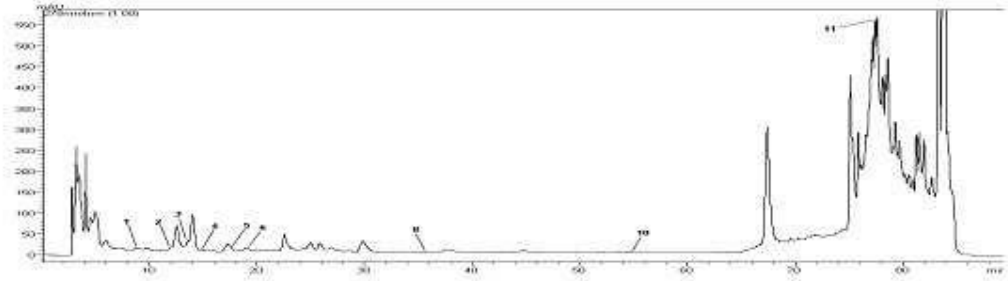

(e)

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Figure 1. HPLC chromatograms of the extracts from *Ganoderma* species (a: Standard phenolics, b: GAW, c: GAM, d: GRW and e: GRM)

[1. Protocatechuic acid, 2. (+)-Catechin, 3. *p*-Hydroxybenzoic acid, 4. Chlorogenic acid, 5. Caffeic acid, 6. (-)-Epicatechin, 7. Ferulic acid, 8. Benzoic acid, 9. Rutin, 10. Rosmarinic acid, 11. Apigenin]

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